

EMAIL  
tomjmorrish@gmail.com  
MOBILE  
+447583588431

# THOMAS JOE MORRISH

---

## EDUCATION

---

### University of Warwick

*PhD in Statistics*

Coventry, UK

2021 - present

- Researching statistical applications of rough paths, specifically building robust inference strategies for discretely observed, multivariate and irreducible rough differential equations. Supervised by Prof A. Papavasiliou and Dr J. Brettschneider
- Co-supervised a summer research student with Prof Papavasiliou, working on statistical methods for rough paths

### University of Potsdam, Theoretical Physics

*Visiting Research Student, Ralf Metzler Group*

Potsdam, Germany

Sep 2025 –

Nov 2025

- Ongoing collaboration includes interdisciplinary methods for anomalous diffusions

### Bocconi University

*MSc in Finance with Quantitative Finance Specialisation*

Milan, Italy

2018 - 2020

- Awarded the **full tuition scholarship**, one of only a few students in the university
- Thesis research included the fast approximation of Fourier inversions for the purposes of pricing vanilla options and extracting beta risk in equities. Manipulation and optimisation over large datasets implemented in MATLAB

### University of Bath

*First Class, BSc (Hons) in Mathematical Sciences with Placement*

Bath, UK

2013 - 2017

- Modules in Pure Mathematics included:
  - o Analysis (88%), Algebra (77%), Group Theory (68%), Projective Geometry (76%)
- Modules in Applied Mathematics included:
  - o Methods and Applications (82%), Probability and Statistics 1 (86%), Modelling and Dynamical Systems (85%), Stochastic Processes and Finance (85%)

### Balcarras School and VI<sup>th</sup> Form

*A Levels: A\*A\*AA in Maths, Further Maths, Economics, Physics (resp.)*

Cheltenham, UK

2006 - 2013

- Advanced Extension Award in Mathematics: Distinction
- Winner of 'Service and Sincerity Award'

---

## EXPERIENCE

---

### Research Assistant

*CAMaCS, University of Warwick*

June 2024 – November 2024

- In collaboration with Cytecom, a biotech startup developing a novel approach to diagnosing bacterial infections.
- Developed signature-based, signal-processing methods to infer cell health from membrane potential dynamics
- Created a synthetic benchmark dataset via a stochastic FitzHugh–Nagumo model to enable robustness testing and algorithm comparison

### Graduate Teaching Assistant

*University of Warwick*

January 2022 – May 2025

- Teaching modules to undergraduate students, including topics in Probability, Statistics and Statistical Computing
- Nominated twice for the department teaching award

### Research/Teaching Assistant

*Università Bocconi*

June 2020 – September 2020

- Researching topics in interest rate modelling
  - Investigating the uses of Ruby on Rails webpage applications
  - Implementing a Kalman Filter approach to signal decomposition in financial time series
  - Writing several exams and building slide packs for teaching
-

---

## FURTHER EXPERIENCE

---

### Technical assistant

NHS England, North and South  
Gloucester PCN  
January 2021-September 2021

- Responsible for technical tasks involved in the administering of the covid vaccine to the population of Gloucester
- Built an excel tool for managing the end-of-life care of vulnerable patients
- Gave instructional classes on the basics of managing data in Excel

### Mathematics Tutor

Jan 2018 – Jan 2020

- Taught GCSE's and A Levels to 12 children simultaneously
- Levels taught ranged from struggling Year 7 pupils to excelling A level students preparing for STEP exams
- Performance of students above expectations across the board
- 200+ hours of teaching experience

### Visiting Research Consultant

PHASE Nepal (NGO), Kathmandu  
December 2017

- Delivered lectures on the uses of Excel and VBA to approximately 50 employees
- Worked on an international research project between Médecins du Monde and PHASE

### Matching Solutions Analyst

LGIM  
June 2015 – July 2016

- Asset/Liability risk and return modelling
- Analysing the effect of tail risk scenarios on DB schemes
- Modelled bespoke assets using regression analysis
- Helped to develop the platform for optimising schemes' liability-aware cashflow matching portfolios

### Junior Chef

Le Champignon Sauvage  
July 2010

Role in 2 Michelin star restaurant included:

- Working closely with head chef, starter chef and pastry chef

---

## PUBLICATIONS, TALKS AND POSTERS

---

- Preprint: *The Inverse Problem for Single Trajectories of Rough Differential Equations* (Thomas Morrish, Anastasia Papavasiliou, Theodore Papamarkou, Yang Zhao). Available at: <https://arxiv.org/abs/2201.10300>
- Talk at the Isaac Newton Institute, Cambridge, for the "Modelling and Applications of Anomalous Diffusions" workshop. Talk available here: <https://youtu.be/dTcQAzCEOfE?si=e-YXlnNE2cotdDYy>
- Talk for the Statistics Summer Interns in the Warwick Stats Department, Summer 2022
- Talk at the Metzler Group Seminar, Autumn 2025
- Posters at the 3rd and 4th BioInference conference, University of Warwick, 2024, and Bardonecchia, 2025
- Talk at 47th Research Students Conference (RSC) in Probability and Statistics, University of Exeter, 2024
- Poster at 46th RSC in Probability and Statistics, University of Sheffield, 2023
- Attended and given talks at several Warwick-based and external reading groups, including Population Genetics, Computing for the Natural Sciences, Philosophy of Modelling.

---

## SKILLS AND ACHIEVEMENTS

---

- **GMAT:** 760/800 (top 1% of candidates)
- Gold and Bronze Duke of Edinburgh awards
- Built an interactive android app in JAVA and XML

### Programming Languages

Python, VBA, MATLAB, R, JAVA

### Languages

**Intermediate:** Italian **Beginner:** German

---

## INTERESTS

---

I have played the violin to a grade 5 standard, and have enjoyed learning how to make ceramics objects, such as ramen bowls and pottery sculptures. I have played cricket, tennis, golf, badminton competitively, and played many more sports socially. I maintain a love of food, cooking at home whenever possible; I also built a full brick pizza oven with no construction experience. Otherwise, I enjoy walking my *lively* 1-year-old dog, Louis.

---